

- 5 Vertically, from the starting point to the highest point,

$$v_y^2 = u_y^2 + 2a_y s_y$$

$$0 = u_y^2 + 2(-9.81)(210)$$

$$u_y = 64.2 \text{ m s}^{-1}$$

$$v_y = u_y + a_y t$$

$$0 = 64.2 - 9.81t$$

$$t = 6.54 \text{ s}$$

Horizontally, across the whole path,

$$s_x = u_x t$$

$$500 = u_x (6.54 \times 2)$$

$$u_x = 38.2 \text{ m s}^{-1}$$

Hence, calculating the angle of the initial velocity,

$$\theta = \tan^{-1} \left(\frac{64.2}{38.2} \right) = 59.2^\circ$$

Answer: D

- 6 By conservation of linear momentum